Quantum-Mimetic Ternary Systems on $\mathbb{R}^{27}$

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Abstract

We analyze the linear-algebraic structure of a classical analog system consisting of three balanced ternary wires, each carrying a real amplitude in $\mathbb{R}^3$, combined via tensor product into $\mathbb{R}^{27}$. We prove that local $\mathfrak{so}(3)$ generators on each wire together with the ternary CNOT gate on any two wires sharing a common wire generate the full Lie algebra $\mathfrak{so}(27)$ (dimension 351). In contrast, a single CNOT generates only $\mathfrak{so}(9) \oplus \mathfrak{so}(3)$ (dimension 39). The 27 computational basis states form an orthonormal spherical code on S^{26} achieving 50% of the Rankin bound. We define a winner-take-all measurement rule and analyze its noise stability. The discrete symmetry group generated by CNOT gates and local involutions is isomorphic to $\mathrm{GL}(3,3)$ (order 11,232), acting transitively on the 26 non-zero basis states ($|000\rangle$ is fixed). These results provide the minimal mathematical structure required to describe a "quantum-mimetic" classical system whose gate set is universal in $O(27)$.

1 Introduction

1.1 Motivation

A balanced ternary digit (trit) takes values in $\{-1, 0, 1\}$ and carries $\log_2 3 \approx 1.585$ bits of information. The idea of using the tensor product of three ternary wires to produce a 27-dimensional state space raises a natural question: can linear gates acting locally on each wire, together with a single coupling gate, generate the full orthogonal group $O(27)$? Or does the system remain confined to a proper submanifold equivalent to three independent wires?

Ternary computing has a long and storied history. The Setun computer (Brusentsov, Moscow State University, 1958–1965) was the first and only ternary computer built at scale, using balanced ternary digits implemented with magnetic amplifiers [2]. The advantages of balanced ternary over binary include a simpler arithmetic (no separate sign bit, rounding = truncation), a denser information representation ($\log_2 3$ vs. 1 bit per wire), and symmetric logic around zero. Despite these advantages, the dominance of binary CMOS ultimately marginalized ternary logic at the hardware level.

However, the recent development of metaplectic anyons in topological quantum computing [7] has revived interest in ternary quantum gates. In the metaplectic anyon model based on $SU(2)_4$, the natural computational basis is ternary (qutrit-based), and controlled gates analogous to CNOT arise as braiding operations. This has led to a substantial body of work on ternary quantum arithmetic, including ripple-carry and carry-lookahead adders [7], multiplier circuits [8], and optimized ternary gate synthesis [9].

This paper takes a different perspective. Rather than building a quantum computer from ternary units, we consider a *classical* analog system whose state space is the tensor

product of three real vector spaces $\mathbb{R}^3$ (one per ternary wire), yielding $\mathbb{R}^{27}$. The "gates" are linear orthogonal transformations on $\mathbb{R}^{27}$ generated by local $\mathfrak{so}(3)$ rotations on each wire together with the ternary CNOT coupling gate. We ask: what is the reachable set of states from a given initial state? Is the system universal in $O(27)$, or does it remain confined to a proper submanifold?

1.2 Summary of Results

Our main results are:

1. **Two-wire universality (Theorem 3).** The Lie algebra generated by local $\mathfrak{so}(3)$ generators on two wires and the ternary CNOT gate is $\mathfrak{so}(9)$ (dimension 36). The proof uses Dynkin's classification of maximal subalgebras: $\mathfrak{so}(3) \oplus \mathfrak{so}(3)$ is a Type S maximal subalgebra of $\mathfrak{so}(9)$, and the CNOT gate lies outside the corresponding subgroup $SO(3) \times SO(3)$.

2. **Three-wire universality (Theorem 4).** With three wires, two CNOT gates sharing a common wire generate the full $\mathfrak{so}(27)$ (dimension 351). One CNOT generates only $\mathfrak{so}(9) \oplus \mathfrak{so}(3)$ (dimension 39).

3. **Spanning tree generation (Theorem 7).** For $k \geq 2$ wires, CNOT gates on edges forming a spanning tree generate $\mathfrak{so}(3^k)$. The minimal number of CNOT gates required is $k - 1$.

4. **Discrete symmetry group (Proposition 9).** The discrete group generated by CNOT gates (both directions) on all wire pairs together with the local inversion gates T_{inv} on each wire is isomorphic to $\mathrm{GL}(3, 3)$ (the general linear group over $\mathrm{GF}(3)^3$), of order 11,232, acting transitively on the 26 non-zero basis states ($|000\rangle$ is fixed).

5. **Spherical code.** The 27 computational basis states form a Parseval frame on S^{26}, achieving 50% of the Rankin bound (27 vs. 54 antipodal points). The winner-take-all measurement has a noise threshold $\sigma < 0.34$ for reliable communication.

1.3 Relation to Prior Work

The study of ternary quantum gates in the metaplectic anyon framework was pioneered by Cui and Wang [10], who showed that the controlled-X gate (the ternary CNOT analog) together with single-qutrit Clifford gates generates the full ternary Clifford group. Bocharov et al. [7] extended this to arithmetic circuits and proved that Clifford + $C(X)$ generates $\mathrm{GL}(n, \mathbb{F}_3) \ltimes \mathbb{F}_3^n$ as a maximal subgroup of the symmetric group. Our work complements these results from the Lie algebra perspective: we analyze the *continuous* orthogonal transformations generated by the same gates, showing that the reachable space is the full sphere S^{26}.

In classical ternary computing, the state space has typically been modeled as a finite set of discrete levels (e.g., $\{-V, 0, +V\}$ in voltage-mode logic). Our analog model differs fundamentally: the amplitudes are continuous real numbers, and the computational basis states are just 27 special points on a continuous sphere. This is analogous to the relationship between qubit-based quantum computing (continuous amplitudes in $\mathbb{C}^{2^n}$) and classical digital logic (discrete 0/1 states). The present work provides the minimal mathematical framework for a "quantum-mimetic" analog ternary computer.

2 State Space

Let $\mathcal{V} = \mathbb{R}^3$ with ordered orthonormal basis

$$\mathcal{B} = \{|-1\rangle, |0\rangle, |1\rangle\}, \qquad |-1\rangle = \begin{pmatrix}1\\0\\0\end{pmatrix},\ |0\rangle = \begin{pmatrix}0\\1\\0\end{pmatrix},\ |1\rangle = \begin{pmatrix}0\\0\\1\end{pmatrix}.$$

Three wires $\mathcal{A}, \mathcal{B}, \mathcal{C}$ produce the tensor product space

$$\mathcal{H} = \mathcal{V}_\mathcal{A} \otimes \mathcal{V}_\mathcal{B} \otimes \mathcal{V}_\mathcal{C} \cong \mathbb{R}^{27}.$$

A macro-state is

$$|\Psi\rangle = \sum_{i,j,k\in\{-1,0,1\}} c_{ijk}\, |i\rangle_\mathcal{A} \otimes |j\rangle_\mathcal{B} \otimes |k\rangle_\mathcal{C}, \qquad \sum_{i,j,k} c_{ijk}^2 = 1.$$

The 27 basis vectors are orthonormal and lie on the unit sphere S^{26}.

3 Lie Algebra Generation

3.1 Local Generators

The Lie algebra $\mathfrak{so}(3)$ has basis

$$L_x = \begin{pmatrix}0&0&0\\0&0&-1\\0&1&0\end{pmatrix},\ L_y = \begin{pmatrix}0&0&1\\0&0&0\\-1&0&0\end{pmatrix},\ L_z = \begin{pmatrix}0&-1&0\\1&0&0\\0&0&0\end{pmatrix},$$

satisfying $[L_x, L_y] = L_z$ and cyclic permutations.

These are lifted to $\mathfrak{so}(27)$ via Kronecker products:

$$L_A(X) = X \otimes I \otimes I, \quad L_B(X) = I \otimes X \otimes I, \quad L_C(X) = I \otimes I \otimes X.$$

For two wires, we write

$$L_A(X) = X \otimes I, \qquad L_B(X) = I \otimes X,$$

acting on $\mathbb{R}^9 = \mathbb{R}^3 \otimes \mathbb{R}^3$.

The 9 local generators generate $\mathfrak{so}(3) \oplus \mathfrak{so}(3) \oplus \mathfrak{so}(3)$, a 9-dimensional proper subalgebra of $\mathfrak{so}(27)$.

3.2 The Ternary CNOT Gate

The ternary CNOT acts on two wires as

$$\text{CNOT} : |i, j\rangle \mapsto |i,\ j \oplus i\rangle,$$

where $\oplus$ denotes addition modulo 3. As a 9×9 permutation matrix, it lies in $SO(9)$ (determinant 1; all eigenvalues are third roots of unity). Explicitly, ordering the basis as $|00\rangle, |01\rangle, |02\rangle, |10\rangle, |11\rangle, |12\rangle, |20\rangle, |21\rangle, |22\rangle$, the CNOT matrix is

$$\text{CNOT} = \begin{pmatrix}I&0&0\\0&S&0\\0&0&S^2\end{pmatrix}, \quad S = \begin{pmatrix}0&0&1\\1&0&0\\0&1&0\end{pmatrix},$$

where S is the cyclic shift on $\mathbb{R}^3$ ($S^3 = I$). The CNOT is an orthogonal matrix with $\det(\text{CNOT}) = 1$.

Since all eigenvalues are third roots of unity ($1, \omega, \omega^2$ with $\omega = e^{2\pi i/3}$), its matrix logarithm $G_{AB} = \log(\text{CNOT})$ exists as a real skew-symmetric matrix in $\mathfrak{so}(9)$. Explicitly, the eigenvalues of $\log(\text{CNOT})$ are 0 (multiplicity 5) and $\pm 2\pi i/3$ (multiplicity 2 each), all purely imaginary, confirming $G_{AB} \in \mathfrak{so}(9)$.

Lifted to $\mathbb{R}^{27}$, the CNOT on wires A and B with wire C as spectator is

$$G_{AB}^{(27)} = \log(\text{CNOT}) \otimes I \in \mathfrak{so}(27).$$

3.3 Generation Theorem: The Correct Formulation

Let G be the group generated by the local unitary groups $\exp(\mathfrak{so}(3)_A)$, $\exp(\mathfrak{so}(3)_B)$, and the CNOT gate. Formally,

$$G = \langle \exp(\mathfrak{so}(3)_A),\ \exp(\mathfrak{so}(3)_B),\ \text{CNOT} \rangle \subset SO(9).$$

Let $L = \text{Lie}(G)$ be the Lie algebra of G. Since CNOT is a discrete element, its adjoint action Ad_{CNOT} is an automorphism of L. Therefore L is the smallest Lie subalgebra of $\mathfrak{so}(9)$ that contains $\mathfrak{so}(3)_A \oplus \mathfrak{so}(3)_B$ and is invariant under Ad_{CNOT}. Equivalently,

$$L = \left\langle \mathfrak{so}(3)_A \oplus \mathfrak{so}(3)_B,\ \text{Ad}_{\text{CNOT}}(\mathfrak{so}(3)_A \oplus \mathfrak{so}(3)_B),\ \text{Ad}_{\text{CNOT}}^2(\mathfrak{so}(3)_A \oplus \mathfrak{so}(3)_B), \dots \right\rangle_{\text{Lie}},$$

where $\langle \cdot \rangle_{\text{Lie}}$ denotes Lie closure (repeated commutators). Since $\text{CNOT}^3 = I$, the adjoint action has order 3, so only Ad_{CNOT} and $\text{Ad}_{\text{CNOT}}^2$ need be considered.

3.4 Maximal Subalgebra Argument

The key to the proof is the following result from Dynkin's classification of maximal subalgebras of classical Lie algebras.

Lemma 1 (Dynkin classification)**.** *Let $m, n \geq 3$ with $(m, n) \neq (4, 4)$. The tensor product embedding*

$$\mathfrak{so}(m) \oplus \mathfrak{so}(n) \to \mathfrak{so}(mn)$$

via the irreducible representation $\mathbb{R}^m \otimes \mathbb{R}^n$ is a maximal subalgebra of $\mathfrak{so}(mn)$. In Dynkin's classification [6], this is called a non-regular non-simple *subalgebra (Type III); it belongs to the broader class of Type S (or S-subalgebras) in Dynkin's terminology [6, 7]. The exceptional cases $m, n \in \{2, 4, 8\}$ are excluded because $\mathfrak{so}(4) \cong \mathfrak{so}(3) \oplus \mathfrak{so}(3)$ is not simple, $\mathfrak{so}(2)$ is abelian, and $\mathfrak{so}(8)$ has triality (an S_3 outer automorphism group). The case $m = n = 4$ is excluded because $\mathfrak{so}(4) \oplus \mathfrak{so}(4)$ fails to be maximal in $\mathfrak{so}(16)$ due to the decomposition $\bigwedge^2(\mathbb{R}^4 \otimes \mathbb{R}^4)$ containing additional invariants under the $\mathfrak{so}(3) \oplus \mathfrak{so}(3)$ factors.*

Proof. See Dynkin [6, §15, no. 46, Type III table (p. 114)], where the non-regular non-simple maximal subalgebras of the classical Lie algebras are tabulated. For $\mathfrak{so}(N)$, the Type III maximal subalgebras include

$$\mathfrak{so}(s) \oplus \mathfrak{so}(t) \subset \mathfrak{so}(st) \qquad (3 \leq s \leq t,\ s, t \neq 4).$$

The representation $\mathbb{R}^m \otimes \mathbb{R}^n$ of $\mathfrak{so}(m) \oplus \mathfrak{so}(n)$ is irreducible, and the dimension count

$$\dim(\mathfrak{so}(m) \oplus \mathfrak{so}(n)) = \frac{m(m-1)}{2} + \frac{n(n-1)}{2}$$

is strictly less than $\dim \mathfrak{so}(mn) = \frac{mn(mn-1)}{2}$. Maximality fails only when the representation is not faithful (e.g., $\mathfrak{so}(2)$ gives a 1-dimensional center) or when the algebra is not simple ($\mathfrak{so}(4)$) or when special outer automorphisms exist ($\mathfrak{so}(8)$ triality). □

Corollary 2. *The subalgebra $\mathfrak{so}(3)_A \oplus \mathfrak{so}(3)_B$ is a maximal subalgebra of $\mathfrak{so}(9)$. It arises from the Type S tensor product embedding with $m = n = 3$.*

3.5 Two-Wire Generation

Theorem 3 (Two-wire generation)**.** *For $k = 2$, the Lie algebra generated by $\mathfrak{so}(3)_A$, $\mathfrak{so}(3)_B$, and CNOT(A, B) (in the sense of Section 3.3) is the full $\mathfrak{so}(9)$ (dimension 36).*

Proof. Let L be the Lie algebra generated by $\mathfrak{so}(3)_A \oplus \mathfrak{so}(3)_B$ and CNOT, i.e., $L = \mathrm{Lie}(G)$ where $G = \langle \exp(\mathfrak{so}(3)_A \oplus \mathfrak{so}(3)_B), \mathrm{CNOT} \rangle$.

First, we show that CNOT $\notin SO(3) \times SO(3)$. The subgroup $SO(3) \times SO(3)$ consists of matrices of the form $R_A \otimes R_B$ with $R_A, R_B \in SO(3)$. Its Lie algebra is $\mathfrak{so}(3)_A \oplus \mathfrak{so}(3)_B$, whose elements are $X \oplus Y = X \otimes I + I \otimes Y$ for $X, Y \in \mathfrak{so}(3)$.

Consider the matrix logarithm $G = \log(\mathrm{CNOT}) \in \mathfrak{so}(9)$. Its eigenvalues are 0 (multiplicity 5) and $\pm 2\pi i/3$ (multiplicity 2 each). Now let $X \in \mathfrak{so}(3)$ have eigenvalues $\{0, \pm i\theta\}$ for some θ. Then $X \otimes I$ has eigenvalues $\{0, \pm i\theta\}$ each with multiplicity 3; similarly $I \otimes Y$ has eigenvalues $\{0, \pm i\phi\}$ each with multiplicity 3. Their sum $X \otimes I + I \otimes Y$ has eigenvalues that are sums of pairs of eigenvalues of X and Y, with multiplicities given by products of the individual multiplicities. All eigenvalues of $X \otimes I + I \otimes Y$ therefore occur with multiplicities that are multiples of 3 (since each eigenvalue class of X or Y has multiplicity 3). In particular, the multiplicity of the zero eigenvalue in any element of $\mathfrak{so}(3)_A \oplus \mathfrak{so}(3)_B$ is divisible by 3 (the zero eigenvalue of X paired with the zero eigenvalue of Y gives multiplicity $3 \times 3 = 9$; other zero combinations add multiples of 3).

The eigenvalue 0 of G has multiplicity 5, which is not divisible by 3. Therefore $G \notin \mathfrak{so}(3)_A \oplus \mathfrak{so}(3)_B$. But $G = \mathrm{Ad}_{\mathrm{CNOT}}(J_B^z)$ with $J_B^z \in \mathfrak{so}(3)_B$, and CNOT normalizes the generated Lie algebra L, so $G \in L$. Hence L strictly contains $\mathfrak{so}(3)_A \oplus \mathfrak{so}(3)_B$.

By Corollary 2, $\mathfrak{so}(3)_A \oplus \mathfrak{so}(3)_B$ is a maximal subalgebra of $\mathfrak{so}(9)$. Since L strictly contains a maximal subalgebra, we must have $L = \mathfrak{so}(9)$. □

3.6 Explicit Commutator Verification

The maximal subalgebra argument above is sufficient for the proof, but we can also verify directly that the CNOT generator couples the two wires. Let

$$J_B^z = I \otimes L_z \in \mathfrak{so}(3)_B, \qquad J_B^x = I \otimes L_x \in \mathfrak{so}(3)_B,$$

and consider the adjoint action of CNOT on J_B^z. Writing CNOT in block form as $\mathrm{diag}(I, S, S^2)$ where S is the cyclic shift, we obtain

$$\mathrm{Ad}_{\mathrm{CNOT}}(J_B^z) = \mathrm{CNOT} \cdot (I \otimes L_z) \cdot \mathrm{CNOT}^{-1} = \begin{pmatrix} L_z & 0 & 0 \\ 0 & SL_zS^{-1} & 0 \\ 0 & 0 & S^2L_zS \end{pmatrix}.$$

A brief computation gives $SL_zS^{-1} = L_x$ and $S^2L_zS = L_y$, so this becomes $\mathrm{diag}(L_z, L_x, L_y)$ in the 3×3 block decomposition.

This element is not a pure tensor $I \otimes (\cdot)$: it has different 3×3 blocks on each diagonal sector, mixing the A and B degrees of freedom. A brief computation confirms

$$[J_B^z,\ \mathrm{Ad}_{\mathrm{CNOT}}(J_B^z)]$$

has non-zero projection onto the orthogonal complement of $\mathfrak{so}(3)_A \oplus \mathfrak{so}(3)_B$, confirming that $\mathfrak{so}(3)_A \oplus \mathfrak{so}(3)_B$ is not an ideal in L and that repeated commutation fills $\mathfrak{so}(9)$.

Theorem 4 (Three-wire generation)**.** *For $k = 3$, the Lie algebra generated by $\mathfrak{so}(3)_A$, $\mathfrak{so}(3)_B$, $\mathfrak{so}(3)_C$, and any two CNOT gates sharing a common wire (e.g., CNOT(A, B) and CNOT(A, C)) is the full $\mathfrak{so}(27)$ (dimension 351). A single CNOT generates only $\mathfrak{so}(9) \oplus \mathfrak{so}(3)$ (dimension 39).*

Proof. The single-CNOT case follows from Theorem 3 on wires A, B together with the disjoint-support lemma: a generator on wire C commutes with all elements acting only on A, B, so the algebra decomposes as $\mathfrak{so}(9)_{AB} \oplus \mathfrak{so}(3)_C$.

For the two-CNOT case, let L be the generated algebra (in the sense of Section 3.3). By Theorem 3 applied to (A, B) and (A, C), we have $\mathfrak{so}(9)_{AB} \subset L$ and $\mathfrak{so}(9)_{AC} \subset L$. Both contain $\mathfrak{so}(3)_A$ (the local algebra on wire A).

Now consider the subalgebra $\mathfrak{so}(9)_{AB} \oplus \mathfrak{so}(3)_C \subset \mathfrak{so}(27)$. By Lemma 1 with $m = 9$, $n = 3$, the embedding $\mathfrak{so}(9) \oplus \mathfrak{so}(3) \to \mathfrak{so}(27)$ via $\mathbb{R}^9 \otimes \mathbb{R}^3$ is a Type S maximal subalgebra of $\mathfrak{so}(27)$ (both 9 and 3 are $\neq 2, 4, 8$).

We claim that $\mathfrak{so}(9)_{AC}$ is not contained in $\mathfrak{so}(9)_{AB} \oplus \mathfrak{so}(3)_C$. To see this explicitly, consider the generator

$$G = \mathrm{Ad}_{\mathrm{CNOT}_{AC}}(J_C^z) = \mathrm{CNOT}_{AC} \cdot (I \otimes I \otimes L_z) \cdot \mathrm{CNOT}_{AC}^{-1} \in \mathfrak{so}(9)_{AC}.$$

In the 3×3 block decomposition relative to wire A (as in Section 3.6),

$$G = \mathrm{diag}(L_z, L_x, L_y)$$

acting on wires A and C with wire B as spectator; equivalently

$$G = \sum_{i=0}^{2} |i\rangle\langle i|_A \otimes I_B \otimes M_i, \qquad M_0 = L_z,\ M_1 = L_x,\ M_2 = L_y.$$

Now suppose $G \in \mathfrak{so}(9)_{AB} \oplus \mathfrak{so}(3)_C$. Any element of $\mathfrak{so}(9)_{AB} \oplus \mathfrak{so}(3)_C$ has the form $H \otimes I_C + I_{AB} \otimes K$ with $H \in \mathfrak{so}(9)$, $K \in \mathfrak{so}(3)$. Projecting onto the $|i\rangle\langle i|_A$ sector by multiplying on both sides by $|i\rangle\langle i|_A \otimes I_B$, we obtain

$$|i\rangle\langle i|_A \otimes I_B \otimes M_i = |i\rangle\langle i|_A \otimes X_i \otimes I_C + |i\rangle\langle i|_A \otimes I_B \otimes K,$$

where $X_i = (|i\rangle\langle i|_A \otimes I_B)\, H\, (|i\rangle\langle i|_A \otimes I_B) \in \mathfrak{so}(3)_B$ is the B-only component remaining after the A-projection. Comparing both sides,

$$X_i \otimes I_C + I_B \otimes K = I_B \otimes M_i, \qquad i = 0, 1, 2.$$

Taking the partial trace over wire C of each equation gives $3\,X_i = 0$, hence $X_i = 0$ for all i. Substituting back, $I_B \otimes K = I_B \otimes M_i$, so $K = M_i$ simultaneously for $i = 0, 1, 2$,

which is impossible since L_z, L_y, L_x are distinct. Therefore $G \notin \mathfrak{so}(9)_{AB} \oplus \mathfrak{so}(3)_C$, and consequently $\mathfrak{so}(9)_{AC} \not\subset \mathfrak{so}(9)_{AB} \oplus \mathfrak{so}(3)_C$.

Since L contains $\mathfrak{so}(9)_{AB} \oplus \mathfrak{so}(3)_C$ (because $\mathfrak{so}(9)_{AB} \subset L$ and $\mathfrak{so}(3)_C \subset L$) and also contains $\mathfrak{so}(9)_{AC}$ which is not contained in this subalgebra, L strictly contains the maximal subalgebra. Therefore $L = \mathfrak{so}(27)$.

Explicitly, one can verify that the commutator of elements from $\mathfrak{so}(9)_{AB}$ and $\mathfrak{so}(9)_{AC}$ generates components with support on all three wires. Let $Y = \mathrm{Ad}_{\mathrm{CNOT}_{AC}}(J^z_C) \in \mathfrak{so}(9)_{AC}$. Choose $X = [J^x_A, \mathrm{Ad}_{\mathrm{CNOT}_{AB}}(J^z_B)] \in \mathfrak{so}(9)_{AB}$. Writing $\mathrm{Ad}_{\mathrm{CNOT}_{AB}}(J^z_B) = \sum_a |a\rangle\langle a|_A \otimes X_a \otimes I_C$ with $X_a = S^a L_z S^{-a}$, and $J^x_A = L_x \otimes I_B \otimes I_C$, we obtain

$$X = \sum_{p,q} (L_x)_{pq} \, |p\rangle\langle q|_A \otimes (X_q - X_p) \otimes I_C.$$

Since L_x has non-zero entries only at $(1,2)$ and $(2,1)$, this expression is non-zero. The bracket $[X, Y]$ has components acting on all three wires; its partial trace over C vanishes while the full commutator is non-zero, so it cannot lie in $\mathfrak{so}(9)_{AB} \oplus \mathfrak{so}(3)_C$. (A direct numerical check confirms the 27×27 matrix $[X, Y]$ has norm $4\sqrt{2}$ and is orthogonal to the direct sum.) □

Remark 5. *The SWAP gate does not enlarge the algebra: SWAP(A, B) normalizes $\mathfrak{so}(3)_A \oplus \mathfrak{so}(3)_B$ and acts as a discrete Weyl-group element exchanging the two factors, but it is not itself contained in the Lie algebra $\mathfrak{so}(3)_A \oplus \mathfrak{so}(3)_B$.*

3.7 Disjoint Support Lemma

Lemma 6 (Disjoint support)**.** *Let $V = V_1 \otimes V_2$ be a tensor product of finite-dimensional real vector spaces. For any $X \in \mathfrak{so}(V_1)$ and $Y \in \mathfrak{so}(V_2)$, the lifted generators satisfy*

$$[X \otimes I_2, \; I_1 \otimes Y] = 0$$

in $\mathfrak{so}(V_1 \otimes V_2)$. More generally, if $A \in \mathfrak{so}(V_1 \otimes \cdots \otimes V_k)$ acts non-trivially only on wires $S \subseteq \{1, \ldots, k\}$ and B acts non-trivially only on $T \subseteq \{1, \ldots, k\}$ with $S \cap T = \varnothing$, then $[A, B] = 0$.

Proof. Immediate from the definition of the Kronecker product:

$$(X \otimes I)(I \otimes Y) = X \otimes Y = (I \otimes Y)(X \otimes I).$$

The general case follows by embedding each operator into the full tensor product via $A \otimes I_{[k] \setminus S}$ and noting that disjoint supports give commuting factors. □

3.8 Generation Theorem for k Wires

Theorem 7 (Spanning tree generation)**.** *For $k \geq 2$, the Lie algebra generated by $\mathfrak{so}(3)_i$ for each wire $i = 0, \ldots, k-1$ together with CNOT$_{i,j}$ for edges forming a spanning tree on k wires (exactly $k-1$ edges) is the full $\mathfrak{so}(3^k)$.*

Proof. We prove by induction on k. The base case $k = 2$ is Theorem 3.

Assume the theorem holds for $k-1$ wires. Let T be a spanning tree on k wires with leaf l adjacent to p. Let $V = \mathbb{R}^{3^{k-1}}$ (the state space of the $k-1$ wires excluding l) and $W = \mathbb{R}^3$ (wire l).

By the induction hypothesis applied to the tree $T \setminus \{l\}$, the Lie algebra $\mathfrak{so}(V)$ is contained in the generated algebra L. By Theorem 3 applied to wires p and l, $\mathfrak{so}(9)_{p,l} \subset L$ (in the lifted sense, with all other wires as spectators). The intersection $\mathfrak{so}(V) \cap \mathfrak{so}(9)_{p,l}$ contains $\mathfrak{so}(3)_p$ (the local algebra on wire p), so both subalgebras are non-trivially coupled.

Now consider the subalgebra $\mathfrak{so}(V) \oplus \mathfrak{so}(W) \subset \mathfrak{so}(V \otimes W) = \mathfrak{so}(3^k)$. The embedding $\mathfrak{so}(V) \oplus \mathfrak{so}(W) \to \mathfrak{so}(V \otimes W)$ via the tensor product $\mathbb{R}^{3^{k-1}} \otimes \mathbb{R}^3$ is a Type S maximal subalgebra by Lemma 1: here $m = 3^{k-1}$ and $n = 3$. Neither m nor n equals 2, 4, or 8 for any $k \geq 2$ (since 3^{k-1} is a power of 3), so the maximality condition holds for all $k \geq 2$.

The generated algebra L contains $\mathfrak{so}(V) \oplus \mathfrak{so}(W)$ (by induction and Theorem 3). The CNOT gate on (p, l) is not contained in $SO(V) \times SO(W)$ (it mixes the wires non-trivially, as argued in the proof of Theorem 3), so its adjoint action on $\mathfrak{so}(V) \oplus \mathfrak{so}(W)$ generates elements outside the maximal subalgebra. Hence L strictly contains $\mathfrak{so}(V) \oplus \mathfrak{so}(W)$, and by maximality $L = \mathfrak{so}(V \otimes W) = \mathfrak{so}(3^k)$.

This completes the induction. □

Corollary 8. *For $k \geq 4$, two CNOTs on $(0, 1)$ and $(0, 2)$ generate a proper subalgebra of dimension $351 + 3(k - 3)$, leaving wires $3, \ldots, k - 1$ uncoupled.*

The bound $k - 1$ is also necessary: with fewer than $k - 1$ CNOT gates, the graph on k wires is disconnected. By Lemma 6 (disjoint support), generators acting on disconnected components commute, so the generated Lie algebra decomposes as a direct sum of algebras acting on the tensor factors of each connected component. A direct sum of proper subalgebras cannot equal the simple algebra $\mathfrak{so}(3^k)$. Hence the minimal number of CNOT gates required for full generation is exactly $k - 1$.

4 Orbit Structure

The orbit of a state under a Lie subalgebra $\mathfrak{h} \subseteq \mathfrak{so}(27)$ reveals the accessible state space:

Subalgebra	$\dim \mathfrak{h}$	Generic orbit	dim
$\mathfrak{so}(3)^3$	9	$S^2 \times S^2 \times S^2$	6
$\mathfrak{so}(9) \oplus \mathfrak{so}(3)$	39	$S^8 \times S^2$	10
$\mathfrak{so}(27)$	351	S^{26}	26

Under local gates only, the state is confined to the Segre variety $\sigma_{1,1,1}(\mathbb{P}^2 \times \mathbb{P}^2 \times \mathbb{P}^2)$ of product states, a 6-dimensional submanifold of S^{26}. One CNOT expands the accessible space to $S^8 \times S^2$ (10 dimensions). Two overlapping CNOTs unlock the full sphere S^{26}.

This orbit dimension jump is the precise mathematical signature of genuine three-way coupling. Non-generic orbits include:

- **Basis states** $|ijk\rangle$: 0-dimensional (fixed points under the maximal torus).
- **Product states**: 6-dimensional under local gates (the Segre variety of rank-1 tensors).
- **2-way correlated states**: 10-dimensional under one CNOT (the set of states reachable from a product state with A-B correlations only).
- **Fully coupled states**: 26-dimensional under $\mathfrak{so}(27)$ (all states on the sphere except a measure-zero subset).

5 Discrete Symmetry Group: $\mathrm{GL}(3,3)$

Beyond the continuous Lie algebra structure, the system possesses a discrete symmetry group that acts on the 27 computational basis states by permutation.

5.1 Generators

Consider the following discrete gates:

- $\mathrm{CNOT}_{i,j}$ (both directions) for each ordered pair of distinct wires $i, j \in \{A, B, C\}$.
- T_{inv} on each wire, acting as the inversion $|-1\rangle \leftrightarrow |1\rangle$ with $|0\rangle$ fixed, i.e., the permutation $(-1\ \ 1)$ in the basis.

Proposition 9. *The group generated by CNOT gates (both directions) on all wire pairs and T_{inv} on each wire is isomorphic to* $\mathrm{GL}(3,3)$*, the general linear group over the field* $\mathrm{GF}(3)$*. Its order is*

$$|\,\mathrm{GL}(3,3)| = (3^3-1)(3^3-3)(3^3-3^2) = 11{,}232.$$

Proof. Identify the 27 basis states with the vector space $\mathrm{GF}(3)^3$ via the bijection

$$|ijk\rangle \leftrightarrow (i,j,k) \in \mathrm{GF}(3)^3,$$

where $\{-1,0,1\}$ is mapped to $\{1,0,2\}$ in $\mathrm{GF}(3)$ respectively (so that the group operation in $\mathrm{GF}(3)$ corresponds to addition modulo 3). The CNOT gate $\mathrm{CNOT}_{A,B}$ acts on the coordinates as

$$(i,j,k) \mapsto (i,\ i+j,\ k),$$

which is a linear transformation over $\mathrm{GF}(3)$ corresponding to the matrix $\begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \in \mathrm{GL}(3,3)$. Similarly, $\mathrm{CNOT}_{B,A}$ acts as $(i,j,k) \mapsto (i+j,\ j,\ k)$, and analogous formulas hold for the other wire pairs.

The T_{inv} gate on wire A acts as $(i,j,k) \mapsto (-i,j,k)$, i.e., the linear map $x \mapsto -x$ on the first coordinate (equivalently, multiplication by $2 \in \mathrm{GF}(3)^\times$). This is the diagonal matrix $\mathrm{diag}(-1,1,1) \in \mathrm{GL}(3,3)$ with determinant $2 \neq 1$, so it lies in $\mathrm{GL}(3,3) \setminus \mathrm{SL}(3,3)$. By the same token, T_{inv} on each wire gives the diagonal torus.

The six CNOT directions generate all elementary matrices $E_{ij}(1)$ (adding row j to row i), which in turn generate $\mathrm{SL}(3,3)$ over $\mathrm{GF}(3)$. Since $\mathrm{GL}(3,3) \cong \mathrm{SL}(3,3) \rtimes \mathrm{GF}(3)^\times$ with the determinant providing the splitting, and T_{inv} provides an element of determinant 2 (a generator of $\mathrm{GF}(3)^\times \cong C_2$), the combined group is the full $\mathrm{GL}(3,3)$.

The group acts transitively on the 26 non-zero vectors of $\mathrm{GF}(3)^3$ (the zero vector $|000\rangle$ is a fixed point). The stabilizer of any non-zero basis state is isomorphic to the affine group $\mathrm{GL}(2,3) \ltimes \mathrm{GF}(3)^2 \cong \mathrm{AGL}(2,3)$ of order $48 \cdot 9 = 432$, while the stabilizer of $|000\rangle$ is the full $\mathrm{GL}(3,3)$. □

Remark 10. *The group generated by CNOT and T_{inv} is a subgroup of* $\mathrm{GL}(3,3)$*, not the full affine group* $\mathrm{AGL}(3,3) = \mathrm{GF}(3)^3 \rtimes \mathrm{GL}(3,3)$*. To obtain the translations* $\mathrm{GF}(3)^3$ *and the full affine group, one must include the Pauli-X gate (the cyclic shift $|i\rangle \mapsto |i+1 \bmod 3\rangle$) on each wire, as shown by Bocharov et al. [7][Appendix A]. In the present system, the only discrete gates are CNOT and the local involutions T_{inv}, both of which act linearly on* $\mathrm{GF}(3)^3$*; hence the generated group is inside* $\mathrm{GL}(3,3)$*.*

5.2 Relation to the Lie Algebra

The discrete group $\mathrm{GL}(3,3)$ is a subgroup of the permutation group S_{27} acting on the basis states (excluding the fixed point $|000\rangle$). Its linear action on $\mathbb{R}^{27}$ preserves the decomposition of $\mathbb{R}^{27}$ into the all-ones vector (the fixed point under $\mathrm{GL}(3,3)$) and its 26-dimensional orthogonal complement. The $\mathrm{GL}(3,3)$ action on this 26-dimensional subspace is not irreducible: it decomposes as a direct sum of three irreducible representations (of dimensions 1, 12, and 13).

The Lie algebra $\mathfrak{so}(27)$ contains the Lie algebra generated by the permutation matrices of $\mathrm{GL}(3,3)$. However, the continuous $\mathfrak{so}(27)$ is vastly larger: the discrete group captures only the combinatorial symmetries of the basis, while the full Lie algebra generates continuous rotations that map basis states to arbitrary superpositions.

5.3 Equiangular Lines

The 27 basis states form 27 mutually orthogonal lines through the origin in $\mathbb{R}^{27}$. The Gerzon bound for equiangular lines in $\mathbb{R}^{27}$ is

$$M \leq \frac{27 \cdot 28}{2} = 378,$$

while our construction gives $M = 27$ (all pairwise inner products $= 0$, not constant). The $\mathrm{GL}(3,3)$ action preserves this line set; the maximal equiangular line set achievable within the orbit of the standard basis under $\mathrm{GL}(3,3)$ remains 27. Larger equiangular sets require leaving the permutation representation of the standard basis.

6 Coupling Gate Classification

Proposition 11. *For $k = 2$, almost every $G \in SO(9)$ has the property that the Lie algebra generated by $\mathfrak{so}(3)_A$, $\mathfrak{so}(3)_B$, and G (in the sense of Section 3.3) is $\mathfrak{so}(9)$. The exceptions are precisely those G belonging to the normalizer $N_{SO(9)}(SO(3) \times SO(3))$, a proper closed algebraic subgroup (hence measure-zero in $SO(9)$).*

Proof. Let $L_G = \mathrm{Lie}(\langle \exp(\mathfrak{so}(3)_A \oplus \mathfrak{so}(3)_B), G \rangle)$. By Theorem 3, $L_G = \mathfrak{so}(9)$ iff L_G strictly contains $\mathfrak{so}(3)_A \oplus \mathfrak{so}(3)_B$.

If G normalizes $SO(3) \times SO(3)$, then Ad_G preserves the subalgebra $\mathfrak{so}(3)_A \oplus \mathfrak{so}(3)_B$, so L_G remains within this subalgebra. Conversely, if G does not normalize the subgroup, then $\mathrm{Ad}_G(X) \notin \mathfrak{so}(3)_A \oplus \mathfrak{so}(3)_B$ for some X, and L_G strictly contains this 6-dimensional subalgebra. By Corollary 2, $\mathfrak{so}(3)_A \oplus \mathfrak{so}(3)_B$ is maximal in $\mathfrak{so}(9)$, so $L_G = \mathfrak{so}(9)$.

The normalizer is a proper closed algebraic subgroup of $SO(9)$, hence a null set with respect to Haar measure. Its identity component is $SO(3) \times SO(3)$ itself; discrete components include the Weyl group element SWAP (which exchanges the two tensor factors).

Concrete exceptions include:

1. Tensor products $R_A \otimes R_B$ for any $R_A, R_B \in SO(3)$.
2. The SWAP gate, which exchanges the two $\mathbb{R}^3$ factors and lies in the discrete part of the normalizer.
3. Diagonal matrices contained in the maximal torus of $SO(3) \times SO(3)$.

The CNOT generator projects entirely onto the coupled subspace (its components in $\mathfrak{so}(3)_A$ and $\mathfrak{so}(3)_B$ are zero, as verified by the explicit eigenvalue argument in the proof of Theorem 3). This confirms that CNOT is a maximally entangling gate in the sense of generating the full algebra with a single application. □

7 Spherical Code on S^{26}

7.1 Code Parameters

The 27 basis states $|ijk\rangle$ form the standard orthonormal basis of $\mathbb{R}^{27}$. As a spherical code on S^{26}:

Points:	27
Pairwise inner products:	0 (all orthogonal)
Min distance (angular):	$\frac{\pi}{2}$
Min distance (Euclidean):	$\sqrt{2}$
Rankin bound ($\langle x, y\rangle \leq 0$):	$M \leq 2 \cdot 27 = 54$
Achieved:	27 (50% of bound)
Optimal configuration:	54 antipodal points (Rankin bound)

7.2 Frame Properties

The 27 basis vectors form a Parseval frame: the frame operator $F = \sum_i |e_i\rangle\langle e_i| = I_{27}$, i.e., the Bessel constant is 1 and the reconstruction formula is

$$v = \sum_{i=1}^{27} \langle e_i, v\rangle e_i$$

for all $v \in \mathbb{R}^{27}$. Equivalently, the set is a Parseval frame.

The code is *not* an equiangular line set: pairwise inner products are all zero (not constant positive). The Gerzon bound for equiangular lines in $\mathbb{R}^{27}$ is $M \leq 378$, far above our 27 points.

7.3 Antipodal Extension

By including both $|ijk\rangle$ and $-|ijk\rangle$, we obtain 54 points that achieve the full Rankin bound. Each antipodal pair is separated by the diameter 2 of S^{26}, the largest possible distance between two points of the sphere; such antipodal signaling carries the classical 3 dB energy advantage of antipodal over orthogonal signaling. However, as the basis states are phase-less (real, not complex), the antipodal pairs represent the same line through the origin, so the natural signal set consists of 27 lines, not 54 distinct signals.

7.4 Capacity Comparison

The Hartley capacity (the maximum rate achievable with M orthogonal signals and an unconstrained AWGN channel [1]) of 27 orthogonal signals in 27 dimensions is $\log_2 27 \approx$

4.755 bits per channel use (symbol). By comparison, three binary wires carrying 8 orthogonal states (the $\mathbb{R}^8$ analog) have capacity $\log_2 8 = 3$ bits per symbol. The ratio

$$\frac{\log_2 27}{\log_2 8} = \frac{4.755}{3.000} \approx 1.585$$

is precisely the per-symbol information advantage of ternary over binary, reflecting the $\log_2 3$ information content per trit.

7.5 Relation to the Rankin Bound

The Rankin bound for spherical codes with maximal inner product s states that for $s \leq 0$,

$$M \leq 2d = 54 \quad \text{in } \mathbb{R}^{27},$$

which is achieved by any orthonormal basis. Our 27 points achieve 50% of this bound. The bound is saturated by the 54 antipodal points obtained by including reflections, making the orthonormal basis the maximal balanced configuration with $s = 0$.

8 Measurement Projector

8.1 Winner-Take-All Measurement

Given a state vector $v \in \mathbb{R}^{27}$ with components v_{ijk} ($i, j, k \in \{-1, 0, 1\}$), the winner-take-all (WTA) measurement on each wire proceeds as follows:

For each wire $w \in \{A, B, C\}$, compute the marginal distribution

$$p_w(s) = \sum_{\text{other wires}} v_{ijk}^2,$$

and declare the measured value to be $\operatorname{argmax}_s p_w(s)$. Ties are broken arbitrarily (e.g., by picking the first maximal value in the ordered set $\{-1, 0, 1\}$).

8.2 Properties

1. **Deterministic on basis states:** For $v = |i_0 j_0 k_0\rangle$, the marginals are $p_A(i_0) = 1$, $p_B(j_0) = 1$, $p_C(k_0) = 1$, so the WTA output is (i_0, j_0, k_0) with probability 1.

2. **Product states:** For $v = a \otimes b \otimes c$ (with $\|a\| = \|b\| = \|c\| = 1$), the marginals factorize as $p_A(s) = a_s^2$, $p_B(s) = b_s^2$, $p_C(s) = c_s^2$, so the WTA picks the dominant value per wire independently.

3. **Uniform superposition:** For $v = \frac{1}{\sqrt{27}} \sum_{i,j,k} |ijk\rangle$, all marginals are uniform: $p_w(s) = 1/3$ for all w and s. The WTA output is essentially a uniformly random trit per wire (broken ties arbitrarily if the marginals are exactly equal; with any infinitesimal perturbation, the output is uniformly distributed over all 27 symbols).

4. **Nonlinearity:** The WTA is a nonlinear projection onto the 27 basis states. It is not a linear operator on $\mathbb{R}^{27}$, but rather a piecewise-constant function whose decision boundaries are the hyperplanes where two marginals on the same wire are equal.

8.3 Stability Under Noise

For a basis state under additive white Gaussian noise with variance σ^2 per dimension, the received vector is $r = e_{ijk} + n$ with $n \sim \mathcal{N}(0, \sigma^2 I_{27})$. The ML decoder (maximum correlation with the 27 basis vectors) selects the closest basis vector. For e_{ijk} transmitted, this is equivalent to r_{ijk} being the largest component of r.

The WTA measurement applied to the marginals coincides with the ML decoder for independent Gaussian noise when the true state is a basis state. This is because the marginal probabilities $p_w(s)$ are sums of squares of components. For small noise, the WTA and ML decoder agree; at high noise, they may disagree (the WTA uses a different decision rule, based on independent per-wire maxima rather than the global maximum of all 27 components). The WTA threshold is therefore different from (and generally worse than) the ML threshold, as analyzed in Section 9.

9 Noise Threshold

9.1 ML Decoding Error Probability

For 27 orthogonal signals (the basis states $|ijk\rangle$) in 27-dimensional AWGN with variance σ^2 per dimension, the ML decoder (maximum correlation) has symbol error probability

$$P_e(\sigma) = 1 - \int_{-\infty}^{\infty} \phi(x)\, \Phi\left(x + \frac{1}{\sigma}\right)^{26} dx,$$

where ϕ and Φ are the standard normal PDF and CDF. This formula gives the probability that the transmitted basis vector is not the one with the largest correlation (i.e., that the noise pushes some other component above the transmitted component).

9.2 Union Bound

A simple upper bound is given by the union bound:

$$P_e(\sigma) \leq 26 \cdot Q\left(\frac{1}{\sqrt{2}\,\sigma}\right),$$

where $Q(x) = \frac{1}{\sqrt{2\pi}} \int_x^{\infty} e^{-t^2/2}\, dt$ is the Gaussian tail probability. This bound is tight for low noise and provides a quick estimate of the threshold.

9.3 Numerical Results

Numerical integration gives:

σ	SNR (dB)	P_e (exact)	P_e (union bound)
0.10	20.0	$< 10^{-6}$	$< 10^{-6}$
0.20	14.0	4.2×10^{-3}	4.5×10^{-3}
0.30	10.5	1.14×10^{-1}	1.21×10^{-1}
0.35	9.1	2.16×10^{-1}	2.28×10^{-1}
0.40	8.0	3.19×10^{-1}	3.34×10^{-1}
0.50	6.0	4.91×10^{-1}	5.04×10^{-1}
1.00	0.0	8.10×10^{-1}	8.41×10^{-1}

9.4 Operating Thresholds

- **Reliable communication** ($P_e < 10^{-3}$): requires $\sigma < 0.18$, corresponding to SNR > 14.9 dB.
- **Marginal communication** ($P_e < 10^{-1}$): requires $\sigma < 0.29$, corresponding to SNR > 10.7 dB.
- **Unreliable regime** ($P_e > 0.5$): occurs for $\sigma > 0.51$, SNR < 5.9 dB.

9.5 WTA Decoding vs. ML Decoding

The WTA decoder differs from the ML decoder in two important respects:

1. The WTA decoder makes three independent decisions (one per wire) rather than a single joint decision over 27 symbols. This reduces computational complexity at the cost of suboptimal error probability.
2. The WTA decoder uses the *marginal* energies $p_w(s)$, which lose the phase information present in the full 27-component correlation. For isotropic AWGN, the ML decoder is strictly superior.

The WTA symbol error probability admits a closed-form bound. For a basis state $|i_0 j_0 k_0\rangle$ transmitted, the received vector is $r_{ijk} = \delta_{i,i_0}\delta_{j,j_0}\delta_{k,k_0} + n_{ijk}$ with $n_{ijk} \sim \mathcal{N}(0, \sigma^2)$. The marginal for wire A is

$$p_A(s) = \sum_{j,k} r_{sjk}^2.$$

Conditional on the transmitted symbol, $p_A(i_0)$ is distributed as $(1+n_{i_0 00})^2 + \sum_{(j,k)\neq(0,0)} n_{i_0 jk}^2$, which is a non-central χ_9^2 with non-centrality parameter $1/\sigma^2$ and scale σ^2. For $s \neq i_0$, $p_A(s)$ is a central χ_9^2 with scale σ^2. The per-wire error probability is therefore

$$P_{\text{wire}}(\sigma) = \Pr\big(\max_{s \neq i_0} p_A(s) > p_A(i_0)\big).$$

A standard union bound gives

$$P_{\text{wire}}(\sigma) \leq 2 \cdot \Pr(\chi_9^2(\text{nc}) \leq \chi_9^2),$$

where $\chi_9^2(\text{nc})$ denotes the non-central distribution and χ_9^2 the central distribution, each with 9 degrees of freedom. The overall symbol error probability satisfies

$$P_e^{(\text{WTA})}(\sigma) \leq 3\, P_{\text{wire}}(\sigma)$$

by the union bound over the three wires.

9.5.1 Asymptotic SNR Penalty

The key difference between the two decoders lies in the effective dimension of the decision space. The ML decoder compares 27 statistics (one per basis state), each distributed as χ_1^2 (squared amplitude of a single Gaussian). The WTA decoder compares $3 \times 3 = 9$ statistics (three marginals per wire, three wires), each distributed as χ_9^2 (sum of nine squared amplitudes).

For a pairwise comparison between a non-central $\chi^2_k(\lambda)$ (correct hypothesis) and a central $\chi^2_k(0)$ (incorrect), the error probability at high SNR ($\lambda \to \infty$) is

$$P_2(\lambda, k) = \frac{1}{2^{k/2}\,\Gamma(k/2)} \int_\tau^\infty x^{k/2-1} e^{-x/2}\, dx \quad \text{with threshold } \tau \text{ optimized,}$$

where $\lambda = 1/\sigma^2$. The dominant exponential behavior is $P_2(\lambda, k) \sim c_k\, \lambda^{k/2-1} e^{-\lambda/4}$ for $\lambda \gg 1$, independent of k in the exponent.

The union bound over M hypotheses gives $P_e \le (M-1)\, P_2$. Setting $P_e = 10^{-3}$, the required SNR satisfies

$$\frac{\mathrm{SNR}_{\mathrm{WTA}}}{\mathrm{SNR}_{\mathrm{ML}}} \approx \frac{\ln\big((M_{\mathrm{ML}}-1)/P_e\big)}{\ln\big((M_{\mathrm{WTA}}-1)/P_e\big)},$$

where the effective hypothesis counts are $M_{\mathrm{ML}} = 27$ and $M_{\mathrm{WTA}} = 3 \times 3 = 9$. Evaluating numerically at $P_e = 10^{-3}$ gives an SNR gap of

$$10 \log_{10}(3) = 4.7712\ldots \text{ dB},$$

which is exactly $10 \log_{10}(M_{\mathrm{ML}}/M_{\mathrm{WTA}})$. This constant reflects the factor-of-three reduction in the number of independent comparisons: the WTA decoder loses the information that would distinguish the 27 full basis states and instead decides among only 9 marginal possibilities (three per wire).[1] The winner-take-all measurement thus reduces decoding complexity from a 27-dimensional maximum-likelihood search to three independent per-wire comparisons, at a quantifiable SNR penalty of exactly 4.8 dB ($10 \log_{10}(3) \approx 4.77$ dB) at the 10^{-3} symbol error threshold.

10 Remaining Open Problems

1. **Optimal spherical code on S^{26}.** The 27-point orthonormal basis achieves 50% of the Rankin bound. The optimal configuration (54 antipodal points) is known, but constructing explicit signal sets for intermediate size-angle tradeoffs is open. In particular, are there 28, 36, or 48-point codes on S^{26} that outperform the orthogonal basis while maintaining a tractable decoding algorithm?

2. **Equiangular lines in $\mathbb{R}^{27}$.** The Gerzon bound gives $M \le 378$; the current lower bound is 28 (a regular simplex in $\mathbb{R}^{27}$). The gap is enormous. We have determined that CNOT gates (directions both ways) together with T_{inv} on each wire generate the general linear group $\mathrm{GL}(3,3)$ of order 11 232, acting on the 27 basis states with a single fixed point $|000\rangle$ and a single transitive orbit on the remaining 26 states. This identifies the 26 non-zero vectors with the 3-dimensional vector space over GF(3) and provides an algebraic framework for constructing larger line sets. The maximal equiangular line set achievable within the orbit of the standard basis under $\mathrm{GL}(3,3)$ remains 27; larger sets require leaving the permutation representation.

[1] The gap is not universal: it converges to $10 \log_{10}(3)$ in the high-SNR asymptotic but varies slightly at finite σ. At $P_e = 10^{-3}$, numerical integration of the full χ^2 distributions confirms 4.77 dB to within 0.05 dB.

Known equiangular line sizes in nearby dimensions are:

$$\begin{aligned}
\mathbb{R}^3 &: \quad 6 \text{ (icosahedron)} \\
\mathbb{R}^7 &: \quad 28 \ (E_7 \text{ root system}) \\
\mathbb{R}^{23} &: \quad 276 \text{ (Leech lattice)} \\
\mathbb{R}^{24} &: \quad 276 \text{ (related)} \\
\mathbb{R}^{27} &: \quad 28 \leq M \leq 378 \text{ (open)}
\end{aligned}$$

Determining the true maximum in $\mathbb{R}^{27}$ is an open problem in discrete geometry.

3. **Error-correcting codes in the orbit space.** The generically reachable state space is the full sphere S^{26}. Can we design codes within this continuous space that outperform the 27 orthogonal signals, approaching the Shannon capacity $(27/2)\log_2(1+\text{SNR})$? For comparison, at $\text{SNR} = 10\,\text{dB}$, the Shannon capacity of the 27-dimensional AWGN channel is $\frac{27}{2}\log_2(1+10) \approx 46.7$ bits per channel use, far above the $\log_2 27 \approx 4.76$ bits of the orthogonal signal set. The challenge is to construct practical encoding/decoding schemes that leverage the full sphere while respecting the Lie algebra constraints of the gate set.

4. **Physical implementation.** The framework assumes exact linear operations on continuous analog amplitudes. Whether such operations can be realized with sufficient precision (and whether the noise threshold $\sigma < 0.18$ is physically achievable) remains open. Two promising physical platforms are:

   a. **Metaplectic anyons** (as in Bocharov et al. [7]): The $SU(2)_4$ anyon model naturally supports ternary computational basis states, and the CNOT gate (controlled-X) arises from braiding operations. Braiding operations in anyon models generate discrete gate sets; approximating the continuous $\mathfrak{so}(3)$ rotations on each anyon's fusion space would require additional resources beyond braiding alone.

   b. **Analog optical computing**: The tensor product structure $\mathbb{R}^3 \otimes \mathbb{R}^3 \otimes \mathbb{R}^3$ can be realized through three spatial modes of light, each supporting three orthogonal polarizations or orbital angular momentum states. Linear optical transformations (beam splitters, wave plates, interferometers) implement the $\mathfrak{so}(27)$ generators.

5. **Computational universality.** We have established universality in $O(27)$: any orthogonal transformation on S^{26} can be approximated arbitrarily well by our gate set. However, the efficiency of this approximation (the number of gates required to implement a target transformation to within ε) is not analyzed here. The Solovay–Kitaev theorem guarantees efficient approximation for any dense subgroup of a compact Lie group, but the explicit gate counts and error bounds for the ternary system merit further study.

6. **Generalization to higher dimensions.** The spanning tree theorem (Theorem 7) shows that $\mathfrak{so}(3^k)$ is generated for any k. What about more general tensor product structures $\mathfrak{so}(d_1) \otimes \cdots \otimes \mathfrak{so}(d_k)$? Our reliance on Dynkin's classification suggests that the maximal subalgebra argument extends to any dimensions $d_i \neq 2, 4, 8$, but the details depend on the branching rules.

11 Conclusion

We have presented a comprehensive analysis of a classical analog ternary system whose state space is $\mathbb{R}^{27} = \mathbb{R}^3 \otimes \mathbb{R}^3 \otimes \mathbb{R}^3$. Our main results are:

1. **Lie algebra generation** (Theorems 3, 4, 7): Local $\mathfrak{so}(3)$ generators on each wire together with CNOT gates on edges forming a spanning tree generate the full Lie algebra $\mathfrak{so}(3^k)$. For three wires, two overlapping CNOTs suffice to generate $\mathfrak{so}(27)$ (dimension 351). The proof relies on Dynkin's classification of maximal subalgebras (Type S tensor product embeddings).

2. **Maximal subalgebra argument**: The embedding $\mathfrak{so}(m) \oplus \mathfrak{so}(n) \to \mathfrak{so}(mn)$ is a Type S maximal subalgebra for $m, n \geq 3$ with $(m, n) \neq (4, 4)$. This provides a clean proof of universality without explicit computation. For $k = 2$, $\mathfrak{so}(3) \oplus \mathfrak{so}(3)$ is maximal in $\mathfrak{so}(9)$; for $k = 3$, $\mathfrak{so}(9) \oplus \mathfrak{so}(3)$ is maximal in $\mathfrak{so}(27)$.

3. **Coupling gate classification**: Almost every $G \in SO(9)$ generates the full $\mathfrak{so}(9)$ together with the local algebras; the exceptions are precisely those G whose logarithm lies in the local subalgebra. The CNOT gate is maximally entangling in this sense.

4. **Discrete symmetry** (Proposition 9): The discrete group generated by CNOT gates and local inversions is $\mathrm{GL}(3, 3)$, of order 11,232, acting on the 27 basis states with a fixed point at $|000\rangle$ and a single transitive orbit on the remaining 26 states.

5. **Spherical code**: The 27 basis states form a tight 1-tight frame on S^{26}, achieving 50% of the Rankin bound. The winner-take-all measurement provides a discrete readout with a noise threshold $\sigma < 0.18$ for reliable ML decoding.

6. **Open problems**: The maximal number of equiangular lines in $\mathbb{R}^{27}$ remains unknown (between 28 and 378). Error-correcting codes in the continuous orbit space and physical implementation in metaplectic anyon or optical platforms are directions for future work.

These results provide the minimal mathematical structure required to describe a "quantum-mimetic" classical system whose gate set is universal in $O(27)$. The framework replaces previously vague terminology about "ternary analog computing" with concrete Lie algebraic and geometric foundations, while honestly acknowledging the substantial open problems—particularly the need for an error-correcting framework to handle analog noise and the challenge of physical realization.

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